CCS 3202 Assignment 1 Yue Chenghao 22154

Question 1

For $f_1(n) = n^2 + n\log n + 75$. Terms: n^2 $n\log n$ constant 75

As n becomes very large. n^2 dominates $n\log n$. and both dominate constant

so dominant term is n^2

$f_1(n)$ is on the order of n^2 i.e. $f(n) \in O(n^2)$

For $f_2(n) = 2n + n^2$. Terms: 2^n, n^2

exponential functions grow faster than any polynominal when n is large

so dominant term is 2^n.

Therefore $f_2(n) \in O(2^n)$.

For $f_3(n) = n - n^{\frac{3}{4}}$ Terms: n and $n^{\frac{3}{4}}$

for large n, a higher power of n dominates a lower power.

so dominate term is n.

Therefore $f_3(n) \in O(n)$

$2^n > n^2 > n$. So $f_2 > f_1 > f_3$.

Question 2

Algo 1

Step	operation	frequency
Initialization	3	1
while condition check	1	n
loop	4	n
final assignment	2	1

Total : $A_1(n) = 3 + (n+1) + 4n + 2 = 5n + 6$

Higher order term : $5n$ so ~~$O(n$~~ complexity is $O(n)$

Algorithm 2.

it has 2 nested loops.

initialization:

sum=0 i=0 . 2 operations → constant term

Nested loop run n times for each i.

each iteration does

1 condition check. — 1 op

1 update · sum += a[i][j] — about 3 ops

1 increment — 1 op

around 5 operations per iteration, so ≈ 5n per run

Plus j=0 → 1 constant.

inner cost ≈ 5n+1

outer loop repeat n times plus its own increment.

total ≈ n × (5n+2) = $5n^2+2n$

+ extra condition check ≈ $5n^2+3n$

So $T_2(n) \approx 5n^2+3n+2+2+1 = 5n^2+3n+5$

(ii) Algo 1 : $T_1(n) = 5n+6$

dominate term is 5n so complexity is $O(n)$

Alg 2 : $T_2(n) = 5n^2+4n+5$

dominate term is $5n^2$ so complexity is $O(n^2)$

(iii) $n^2 > n$

Algorithm 1 grows linearly with n while Algorithm grows quadratically

So Algorithm is more efficient

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